1 Key Performance Indicators Relating to Workplaces: Configuration

Overview

This document describes the configurations required to calculate and display KPIs relating to shifts and workplaces.

Integration

The shop floor clients AIP and/or AIP2, the Graphic machinery and Line Andon Board provide KPIs relating to shifts/workplaces. By configuring these clients, you can view these KPIs. The calculated KPIs correspond to those also displayed in the Efficiency report and/or the OEE report.

Requirements

	AIP / AIP2	Graphic machinery	Line Andon Board
Chapter relevant to	X	X	X

Execute the database patch *dbp_mde82.hsc* and enable the required configuration in the *Scheduler*.

Enable the configuration in the *INI configuration* if you want to show the key performance indicators in the AIP2.

Enable the upgrade aipkpi if you want the AIP and/or AIP2 to show the key performance indicators.

Enable the upgrade mpark2kpi if you want the graphic machinery to show the key performance indicators.

Scheduler configuration

	AIP / AIP2	Graphic machinery	Line Andon Board
Chapter relevant to	X	X	X

Enter and enable the following entry in the Scheduler to provide for a cyclic calculation of KPIs relating to shifts/workplaces.

Field	Value
Type	S (Standard)
Category	I (interval)
Alterable	Yes
Visible	Visible
Product key	
License key	
Command (Windows)	hymw.exe -u9999 -c“DLG=KEYF.WRITEWPCURSHIFT|RWSC.TIMEOUT=3600|“
Command (Linux)	hymw.out -u9999 -c“DLG=KEYF.WRITEWPCURSHIFT|RWSC.TIMEOUT=3600|“
Comment	MDE key figure calculation
Interval	0:10:00
Active	☑

When you install the MDE version 8.2, the database patch *dbp_mde82.hsc* automatically imports this entry. However, the entry is disabled by default. Please check at first if the entry already exists, before you attempt to enter a new one.

The duration required for the calculation of KPIs depends on the following criteria:
- the workplaces you want to calculate the KPIs for
- the KPIs you want to calculate.

Therefore, you should adjust the interval to the number of workplaces and the number of KPIs you want to calculate. To do so, define the workstations and the KPIs first, then execute the command directly in a server system environment and check the duration.

The system updates the calculated KPIs relating to shifts/workplaces in the table *keyfigures_current_shift*.

INI configuration displaying KPIs in the AIP and AIP2

	AIP / AIP2	Graphic machinery	Line Andon Board
Chapter relevant to	X	-	-

Add the following entry to the INI configuration in order for the AIP and/or AIP2 to show key performance indicators. Then restart HYDRA.

Field	Value
Name	MDE
Section	AIP

Field	Value
Key	KEYFIGURES
Value	Y
Active	☑
Comment	Display key figures in AIP

Processing information about the AIP and AIP2

	AIP / AIP2	Graphic machinery	Line Andon Board
Chapter relevant to	X	-	-

A service cyclically run via the Scheduler calculates KPIs. The calculated KPIs correspond to those also displayed in the Efficiency report and/or the OEE report.

The following cycles specify when data is shown in the AIP or AIP2:

- Server processing: according to Scheduler configuration. By default: 600 seconds.

  The server runtime depends on:
  - the number of KPIs you want to calculate and
  - the number of affected workplaces.

- Updating of the machine list (mnr.lst) due to manual postings: at the latest after 900 seconds (default value).

Therefore, the scrap rate might only be updated and displayed after a delay, for example, if you report part quantities.

Updating of KPIs might also be delayed after a shift change. Reason: by default the job runs every 10 minutes, i.e. in the worst case the application shows KPIs of the previous shift for up to 9 minutes after a shift change.

Processing information on the graphic machinery / Line Andon Board

	AIP / AIP2	Graphic machinery	Line Andon Board
Chapter relevant to	-	X	X

A service cyclically run via the Scheduler calculates KPIs. The calculated KPIs correspond to those also displayed in the Efficiency report and/or the OEE report.

The following cycles specify when data is shown in the graphic machinery:

- Server processing: according to Scheduler configuration. By default: 600 seconds.

  The server runtime depends on:
  - the number of KPIs you want to calculate and
  - the number of affected workplaces.

- Updating of the data displayed in the graphic machinery.

Therefore, the scrap rate might only be updated and displayed after a delay, for example, if you report part quantities.

Updating of KPIs might also be delayed after a shift change. Reason: by default the job runs every 10 minutes.

Overview of available key performance indicators

	AIP / AIP2	Graphic machinery	Line Andon Board
Chapter relevant to	X	X	X

The AIP/AIP2 or the Graphic machinery show the following KPIs relating to workplaces/shifts (if necessary, you might have to configure the GUI). You can calculate these KPIs at regular intervals.

The Line Andon Board only provides the OEE.

KPI	Formula ID used for changes in the formula management and for the configuration of limit values (optional)	Acronym for the KPI in the file mnr.lst	Acronym for the color in the file mnr.lst
Utilization efficiency (rate of capacity utilization)	Rcu	KF.RCU	KF.RCU.COLOR
Allocation efficiency	Ocu	KF.OCU	KF.OCU.COLOR
Techn. efficiency	tec_ef	KF.TEC_EF	KF.TEC_EF.COLOR
Rate	yie_ra	KF.YIE_RA	KF.YIE_RA.COLOR
Scrap ratio	scr_ra	KF.SCR_RA	KF.SCR_RA.COLOR
OEE	oee	KF.OEE	KF.OEE.COLOR
Availability	avail	KF.AVAIL	KF.AVAIL.COLOR
Performance	pf_rat	KF.PF_RAT	KF.PF_RAT.COLOR

KPI	Formula ID used for changes in the formula management and for the configuration of limit values (optional)	Acronym for the KPI in the file mnr.lst	Acronym for the color in the file mnr.lst
Quality	qual	KF.QUAL	KF.QUAL.COLOR
Machine run time	mch_rt	KF.MCH_RT	KF.MCH_RT.COLOR
Actual utilization	act_ut	KF.ACT_UT	KF.ACT_UT.COLOR
Yield utilization	yie_ut	KF.YIE_UT	KF.YIE_UT.COLOR

Configuration of workplaces

	AIP / AIP2	Graphic machinery	Line Andon Board
Chapter relevant to	X	X	X

In general, the system calculates key performance indicators for all workplaces. If you only want to calculate KPIs for specific workplaces, use the Advanced object configuration to define the affected workstations:

Field	Value
Object type	fixed "KEYFIGURE_CURRENT_SHIFT“
Object ID 1	fixed "WORKPLACE“
Object ID 2	<workplace/machine number>
Parameter	fixed "CONSIDER_IN_SCHEDULER“
Parameter value	fixed "TRUE“
Active	☑

The system calculates all configured key performance indicators for the required workstations.

You must configure the production line (machine/workplace of the type "L") for the Line Andon Board.

Configuration of the key performance indicators you want to calculate

	AIP / AIP2	Graphic machinery	Line Andon Board
Chapter relevant to	X	X	X

In general, the system calculates all key performance indicators listed above. If you want to restrict the KPIs, use the Advanced object configuration to specify the KPIs you want to calculate.

Field	Value
Object type	fixed "KEYFIGURE_CURRENT_SHIFT“
Object ID 1	fixed "KEYFIGURE“
Object ID 2	<abbreviation of the KPI>, e.g. "rcu“
Parameter	fixed "CONSIDER_IN_SCHEDULER“
Parameter value	fixed "TRUE“
Active	☑

The system calculates key performance indicators for all workplaces or for all workplaces configured above.

Configuration of limit values

	AIP / AIP2	Graphic machinery	Line Andon Board
Chapter relevant to	X	X	X

To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached.

The graphic machinery and the Line Andon Board also highlight their layouts if limit values are exceeded or not reached.

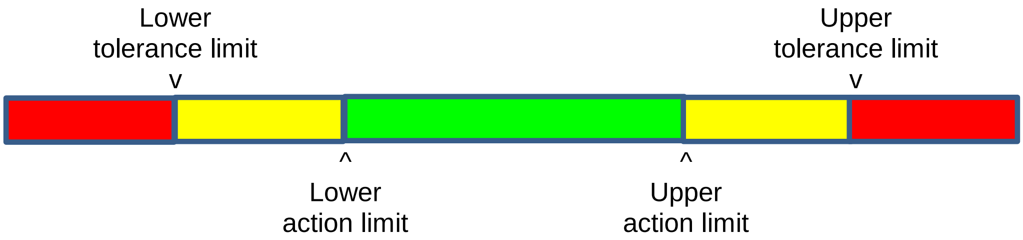
Optionally, you can define workplace-related limit values for color highlighting. You can store these limit values for each KPI and workstation in the database table *keyfigures_target_values.* You must maintain the following fields in the database table:

The table *keyfigures_target_values* is not intended to be maintained via the MOC. You can use a database frontend to edit the data.

Database field	Value
object_type	fixed "WORKPLACE“
object_id1	Workplace number according to configuration.
object_id2	
object_id3	
object_id4	
key_figure	ID of the KPI. Possible values: see below
upper_tolerance_limit	Upper tolerance limit see below
upper_action_limit	Upper action limit see below
upper_tolerance_limit	Lower tolerance limit see below
lower_action_limit	Lower action limit see below
target_value	

Database field	Value
modified_by	Last modified by
modified_ts	Last modified on (date/time)

Colors are identified as follows:



Only set the lower tolerance limit and lower action limit for KPIs aiming at high values (e.g. utilization efficiency/rate of capacity utilization).

Only set the upper tolerance limit and upper action limit for KPIs aiming at low values (e.g. scrap rate).

The value of the KPI is...	Graphic machinery	AIP2
greater than the upper tolerance limit	Red $FF0000	Red $1010D0
between the upper action limit and the upper tolerance limit	Yellow $FFFF00	Orange $1090FF
between the lower action limit and the upper action limit	Green $008000	Green $107000
between the lower action limit and the lower tolerance limit	Yellow $FFFF00	Orange $1090FF
less than the lower tolerance limit	Red $FF0000	Red $1010D0
Other (no limit values configured)		Gray $909090

The server identifies the color. For technical reasons, the color code varies for the graphic machinery and the AIP2.

Configuration of alternative key performance indicators in the AIP2

	AIP / AIP2	Graphic machinery	Line Andon Board
Chapter relevant to	X	-	-

The KPIs displayed by default are configured in the below-mentioned xml files. You must locate the KPIs in the file in order to configure the KPIs. To do so, browse the file for the KPI's abbreviation. For example: <!--"oee" (OEE)-->.

<AIP2 directory>\gui\a_data_mnr.xml → basic screen "tile design“
<!--"oee" (OEE)-->

```
<!--"rcu" (rate of capacity utilization/utilization efficiency)-->

<!--"scr_ra" (scrap rate)-->
```

<AIP2 directory>\gui\a_view_mnr.xml → icon view

```
<!--"oee" (OEE)-->

<!--"rcu" (rate of capacity utilization/utilization efficiency)-->

<!--"scr_ra" (scrap ratio)-->
```

This configuration file defines the format for KPIs not relating to durations. The format specifies that you must enter these key performance indicators with two decimal places:

<AIP2 directory>\gui\globaldefines.xml

```
<!--Format for mde key figures (KF.OEE, KF.RCU, ...; e.g. %g, %0.2f)-->
<FORMAT_MDEKEYFIGURES>%0.2f</FORMAT_MDEKEYFIGURES>
```

In addition to the KPIs displayed by default, you can also configure further KPIs or other workstation/shift-related KPIs to be displayed. The chapter *Overview of available key performance indicators* describes the available KPIs and their IDs (acronyms).

Use the formatting

```
<OnGetDisplayValue>FormatDuration</OnGetDisplayValue>
```

instead of <DisplayFormat...> for KPIs relating to durations. You can find further information on the configuration of the AIP2 in the training documents *Extended Application Training MES Terminal.*

Configuration of KPIs in the AIP basic screen *list form*

	AIP / AIP2	Graphic machinery	Line Andon Board
Chapter relevant to	X	-	-

The basic screens of the AIP and AIP2 can also show KPIs in lists. To do so, configure the relevant settings in the configuration file *ctaiplay.ini*, e.g.:

```
[Maschinenliste]
KF.RCU=N3.2,100,R,rate of capacity utilization/utilization efficiency
KF.MCH_RT=hh:mm,100,Z,run time of machines
```